

SMART ELECTRICAL FAULT REPORTING, CLASSIFICATION & TRACKING SYSTEM

Mini Project Documentation

Department: Electrical & Electronics Engineering (EEE)

Project Type: B.Tech Mini Project

Development Platform: Base44 AI

Academic Year: 2026–2027

Live Application: <https://volt-sense-fix.base44.app>

1. Project Overview

The Smart Electrical Fault Reporting, Classification & Tracking System is a web-based application developed to improve the process of reporting, managing, tracking and analyzing electrical faults. The system provides a centralized platform where users can report faults, maintenance teams can manage them, technicians can update repair progress, and administrators can analyze historical fault information.

2. Problem Statement

Electrical faults in colleges, industries, laboratories, apartments and commercial facilities are often reported through phone calls, messages or manual registers. These methods can cause delayed responses, difficulty prioritizing critical faults, poor tracking and lack of historical maintenance information. A centralized digital system is therefore required to manage the complete electrical fault lifecycle.

3. Proposed Solution

The proposed application provides a structured workflow: User Reports Fault → Fault Classification → Severity Assessment → Technician Assignment → Inspection → Repair → Resolution → Analytics. This converts a traditional complaint-based process into a digital electrical maintenance workflow.

4. Objectives

- Provide an easy platform for reporting electrical faults.
- Maintain centralized fault records.
- Classify faults by electrical category and severity.
- Assign faults to technicians.
- Track faults from reporting to closure.
- Analyze recurring and location-based faults.
- Support preventive maintenance decisions.

5. Target Users

User / Student / Staff: Report faults and track submitted complaints.

Electrical Technician: View assignments, inspect faults, update status and record repair details.

Administrator: Manage users, assign technicians, monitor faults and view analytics.

6. Major Features

Dashboard: Displays total, open, in-progress, resolved and critical faults.

Fault Reporting: Captures fault ID, reporter, location, category, description, date/time, priority and supporting information.

Smart Classification: Organizes faults by type and severity such as Low, Medium, High and Critical.

Technician Assignment: Allows administrators to assign faults to technicians.

Fault Tracking: Provides a status lifecycle from Reported to Closed.

Electrical Analytics: Shows fault trends, categories, locations, resolution time and repeated faults.

Notifications: Supports alerts for assignments, critical faults, status changes and resolution.

7. Fault Categories

The application can support categories including Power Failure, Short Circuit, Overload, Voltage Fluctuation, Electrical Leakage, MCB Tripping, Motor Fault, Lighting Fault, Transformer Fault, Cable Fault and Other.

8. Fault Status Workflow

Reported → Classified → Assigned → Accepted → Inspection → Repair in Progress → Resolved → Closed

9. System Architecture

Users (User / Technician / Admin) interact with the Base44 web application. The application manages fault reporting, technician management and analytics, while application data stores user, fault, assignment, status-history and maintenance information.

10. Technologies Used

- Base44 AI – application development and deployment
- AI prompting – requirements, UI and workflow generation
- Web application technologies – interface and application functionality
- Database – fault and user information
- Authentication – role-based access
- Dashboards and charts – data visualization
- GitHub – documentation and version control

11. Role of AI

Base44 AI was used as an AI-assisted application development platform. Natural-language requirements were provided to generate and refine application pages, forms, dashboards, workflows and data structures. The application was iteratively improved by giving additional prompts based on project requirements.

12. Working Methodology

1. User identifies an electrical problem.
2. User submits a fault report.
3. The system records and classifies the fault.
4. Severity and priority are determined.
5. Administrator assigns a technician.
6. Technician inspects and repairs the fault.
7. Technician updates the status and repair information.
8. The fault is marked resolved and closed.
9. Historical data is analyzed for trends and preventive maintenance.

13. Sample Fault Record

Example: Fault ID EF-001 | Location: EEE Laboratory | Fault Type: MCB Tripping | Severity: High | Status: In Progress | Technician: Technician 01 | Description: Repeated MCB tripping during laboratory operation.

14. Expected Results

The system is expected to reduce manual reporting, improve communication, provide better fault visibility, maintain centralized records, identify frequently occurring faults and support preventive maintenance planning.

15. Advantages

- User-friendly interface
- Centralized fault management
- Faster reporting and assignment
- Status visibility
- Historical maintenance records
- Electrical fault analytics
- Better maintenance planning
- Scalable web-based solution

16. Limitations

The mini-project is primarily a web-based fault management and tracking system. Real-time electrical sensor monitoring would require additional hardware and IoT integration. Advanced prediction accuracy would require a sufficiently large historical dataset and a dedicated machine-learning model.

17. Future Scope

Future versions can integrate IoT sensors for voltage, current, temperature and vibration monitoring; real-time dashboards; AI/ML-based fault prediction; mobile applications; automated critical alerts; GIS-based fault mapping; energy monitoring; and predictive maintenance.

18. Testing

Functional testing should verify fault submission, classification, technician assignment, status updates, fault tracking, resolution, search/filtering and analytics. Each module should produce the expected result for valid user actions.

19. Conclusion

The Smart Electrical Fault Reporting, Classification & Tracking System provides a digital approach to electrical maintenance management. It combines Electrical Engineering concepts, web application development, data management, analytics and AI-assisted development to create a practical academic solution. The system demonstrates how an engineering problem can be transformed into a functional web application using Base44 AI.

20. Project Information

Project Title: Smart Electrical Fault Reporting, Classification & Tracking System

Department: Electrical & Electronics Engineering (EEE)

Development Platform: Base44 AI

Live Demo: <https://volt-sense-fix.base44.app>

GitHub Repository: To be added

Developed By: Student Name(s)

Institution: College Name

Academic Year: 2026–2027

21. Suggested GitHub Repository Structure

Smart-Electrical-Fault-Reporting-System/

■■■■ README.md

■■■■ docs/

■ ■■■■ login.png

■ ■■■■ dashboard.png

■ ■■■■ report-fault.png

■ ■■■■ fault-tracking.png

■ ■■■■ technician-dashboard.png

■ ■■■■ analytics.png

■■■■ documentation/

■ ■■■■ Mini_Project_Report.pdf

■■■■ LICENSE

22. Project Link

Live application: <https://volt-sense-fix.base44.app>

Prepared for academic mini-project documentation and GitHub publication.